Nayoung Kim

HRI Researcher
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ABOUT ME

I am an HRI researcher who translates complex robotic technologies into user-centered interaction experiences. My work focuses on robot service planning, interaction prototyping, and usability evaluation for AI and robotic systems. I aim to bridge the gap between technical possibilities and real user needs through research-driven design.

EDUCATION

Sep 2021 – Aug 2023

Yonsei University, Seoul, Korea, Republic of
- Master of Science in Industrial Engineering
- Major: Human-Computer Interaction (HCI), Advisor: Yong Gu Ji
- GPA: 4.06 / 4.3

Mar 2017 – Aug 2021

Sungshin Women University, Seoul, Korea, Republic of
- Bachelor of Social Science
- Major: Psychology
- GPA: 4.11 / 4.5

WORK EXPERIENCE

Sep 2024 – Present

HRI Lead, AI Robotics Lab, Sogang University, Seoul, Korea, Republic of
(Project 1) Elderly-friendly Shared Autonomy for Teleoperated Robots in Superaged Society
(Project 2) Development of a continual intelligence-enhancing RaaS framework based on human-robot shared control
- Designed LLM-based robot teleoperation interfaces and conducted usability testing for AI robotic systems
- Designed end-to-end interaction flows from early concepts to high-fidelity prototypes
- Collaborated with engineers to align technical development with user-centered design principles
- Conducted psychology-based human-robot interaction research and translated findings into UX requirements

Sep 2023 – Feb 2024

HRI Researcher, AI Robotics Institute, Korea Institute of Science and Technology (KIST)
(Project) Advanced Technology Development Project for Efficient Installation and Operation of Quarantine Treatment Facilities
- Planned AI robotic control services for quarantine treatment facilities and conducted scenario-based usability testing
- Designed psychology-based AI robotic research as part of individual studies and pres ented findings at IROS

Project

Aug 2024 – Present	**Elderly-friendly Shared Autonomy for Teleoperated Robots in Superaged Society** : Research project funded by the National Research Foundation of Korea (NRF) • Planned robot service scenarios through task analysis and field observations • Analyzed user characteristics and developed HRI design guidelines through desk research • Designed wireframes and robot interaction prototypes using Figma and ProtoPie • Identified user needs and pain points through user research and usability testing • Collaborated with social workers, developers, and engineers to translate UX insights into robot system and interface requirements • Submitted research findings to an SCI-indexed international journal (under review)
Aug 2023 – Feb 2024	**Investigating Behavioral and Cognitive Changes Induced by Autonomous Delivery Robots in Incidentally Copresent Persons** : Individual research funded by Korea Institute of Science and Technology (KIST) project of Intelligence on Multi-Robot Autonomy • Planned and validated UX scenarios for autonomous delivery robots • Investigated user perceptions of delivery robot behaviors in shared public spaces • Presented research findings at IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
Aug 2021 – Feb 2022	**Interface Optimization Research for Autonomous Driving UX** : Research project funded by Hyundai Motor Company • Conducted user research to identify pain points and needs in existing infotainment systems • Designed multimodal infotainment interfaces, including navigation, to enhance UX • Performed usability testing and data analysis to validate the enhanced system

PUBLICATIONS

Nov 2025	**Kim, N. Y**., … & Nam, C. J. (2026). Hey, Robot! Don't Cut in Line: Designing Queuing Behaviors of Delivery Robots with Multiple Stakeholders. IEEE Robotics and Automation Letters
Aug 2023	Shin, HR., **Kim, NY.,** … & Ji, Y. G. (2024). Evaluating and eliciting design requirements for an improved user experience in live-streaming commerce interfaces. *Computers in Human Behavior*, *150*, 107990
Apr 2026	Ko, JH., Jung, HC., **Kim, NY.** & Nam, CJ. DEX-Mouse: A Low-cost Portable and Universal Human Interface with Force Feedback for Data Collection of Dexterous Robotic Hands. IEEE Robotics & Automation Letters (Under Review)
Apr 2026	**Kim, NY.**, … & Lee, SO. Beyond Assistance: Self-Motivated Engagement of Older Adults with Robots in Everyday Life. International Journal of Human–Computer Interaction (Under Review)
Nov 2025	Lee, JH., …, **Kim, NY.** & Nam, CJ. HEART: Coordination of Heterogeneous Expert Agents for Physically Grounded Robotic Task Planning. IEEE Robotics & Automation Letters (Under Review)

Nov 2025 **Kim, NY.,** … & Nam, CJ. Mitigating Barriers to Workforce Engagement Associated with Aging: Designing Situation-Aware Interfaces for Teleoperated Robots., International Journal of Social Robotics (Under Review)

Sep 2025 Lee, SB., … **Kim, NY.** & Nam, CJ. Human Factor–Aware Adaptive Task Allocation for Multi-Human Multi-Robot Remote Supervision. IEEE Transactions on Human-Machine Systems (Under Review)

CONFERENCE PRESENTATIONS

2024 **Kim, N.,** & Kwak, S. S. (2024, October). Investigating Behavioral and Cognitive Changes Induced by Autonomous Delivery Robots in Incidentally Copresent Persons. In 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (pp. 2514-2519). IEEE.

2023 **Kim, N.,** … & Kim, S. Analysis of User Experience Requirements for Continuous Use of Digital Healthcare Services for Female Infertility. *PROCEEDINGS OF HCI KOREA 2023*, 402-407

2022 **Kim, N.,** Kim, G., Ji, Y.G., Differences in User Information Perception Depending on the Delivering Modality of the Voice Assistant's Search Results in a Driving Situation., *Proceedings of the ESK Conference 2022*, 284.

PATENT

Dec 2024 **Kim, N.,** Kim, S. "A Medication Management Device Capable of Detecting Emergencies." Korea Patent (10-2747601-0000)

AWARDS

Jun 2022 **Grand Award Smart Pillbox: Elderly Emergency Detection Pillbox for Sustainable Society**
: Smart City Platform Idea Awards, Yonsei University
- Grand Award, Team Work